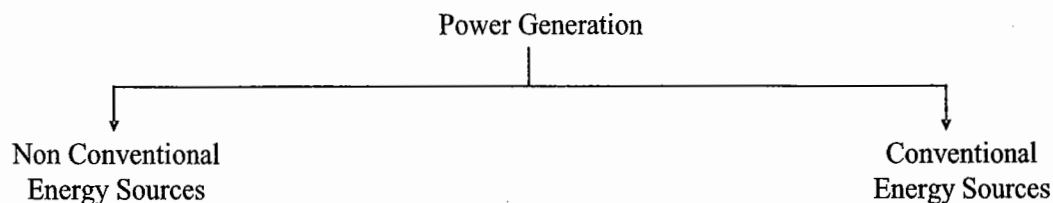


1

Generating Stations



Non Conventional Energy Sources	Conventional Energy Sources
1. Small capacity power Generation over a shorter interval of time Eg:- Wind, Solar, Tidal, Geothermal, Diesel, Biomass and MHD.	1. Bulk capacity power Generation over a longer period. Eg:- Steam (or) Thermal, Hydro, Nuclear and Gas.
2. Most of the plants are located near to consumer premises i.e, there is no constraint in setting up of these plants	2. Most of the plants are located at remote places i.e., these plants are set-up based on geographical constraints.
3. The operation of these plants are based on day-to-day weather constraints	3. There are no weather constraints while operating these plants.
4. Most of these plants having less fixed cost and high running cost.	4. Most of these plants are having high fixed cost and less running cost.
5. Suitable to meet peak load demands except Geothermal plants.	5. Suitable to meet base load demand except Gas plants.
6. Asynchronous Generators are employed Ex:- DC Generator, AC Generator and Induction Generator	6. Synchronous Generations are used
7. Distribution network is sufficient	7. Both the transmission and distribution networks are employed.
8. Generated voltage will be at ≤ 1 kV.	8. Generated voltage will be 3.3, 6.6, 11, 13.2 and 18.2 kV.

1.1 Steam Power Plants

A generating station which converts heat energy of coal combustion into electrical energy is known as a steam power station.

1. Selection of site for steam power stations:

- (a) The site should be nearest to the coal mines so that Transport cost of fuel is minimum.

- (b) Large quantity of cooling (for the condenser) water should be available.
- (c) The land should not be rocky and marshy.
- (d) The chimney of the plant, do not obstruct the flying aero planes, i.e., the site should be away from the airport.
- (e) The site should not be surrounded by residential building to avoid nuisance of smoke, noise etc.
- (f) If the station is located, near the load center, the distribution cost reduces.
- (g) Facility for the disposal of ash.

2. Schematic Arrangement of Steam Power Stations:

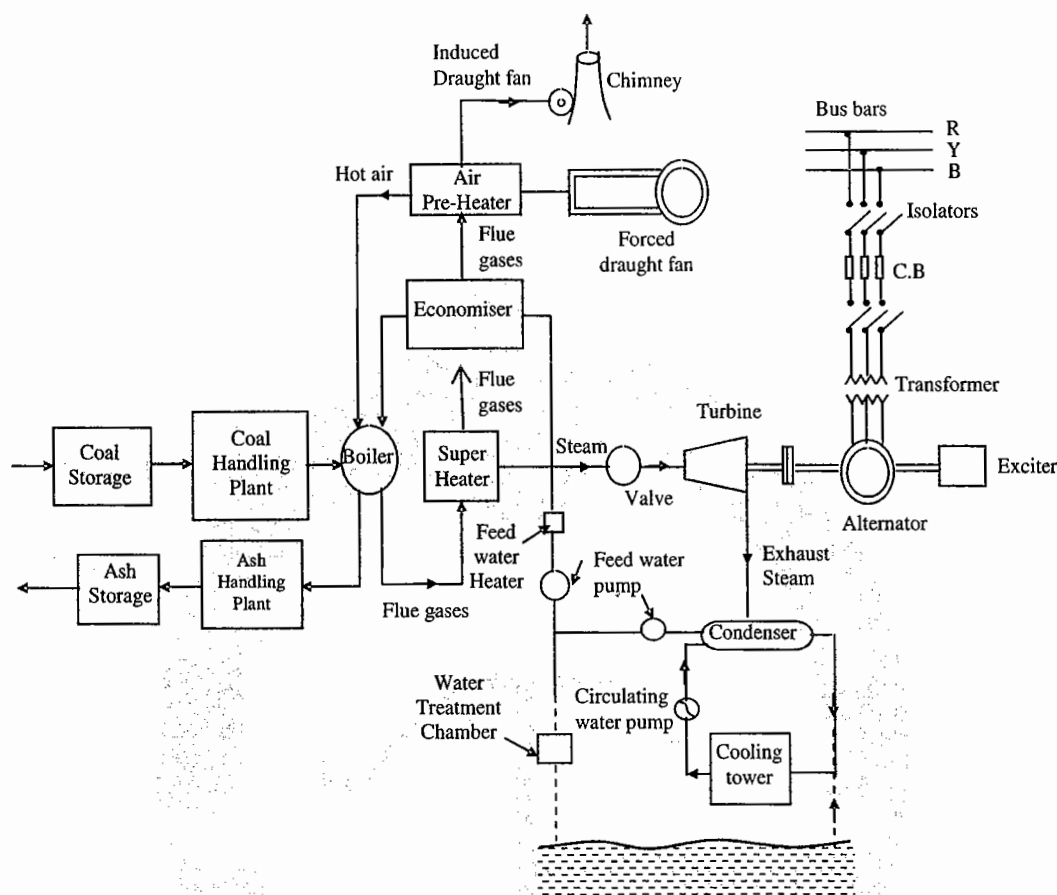


Figure: Schematic diagram of Steam Power stations

3. The main and auxiliary equipment in a thermal power plant are.

- Coal and ash handling plant
- Steam generator or Boiler
- Super heater
- Economizer
- Airpreheater.
- Steam Turbine
- Condenser
- Cooling towers
- Feed water heater
- A. C. Generator (or) Alternator
- Exciter
- Electro static precipitator

(a) Coal and Ash handling plant:

The coal is transported to the power station by road 'or' rail and is stored in the coal storage plant. The coal is delivered to the coal handling plant. Where it is pulverized (i.e., crushed into small pieces) in order to increase its surface exposure. Thus promoting rapid combustion without using large quantity of excess air. The pulverized coal is fed to the boiler by belt conveyors.

Advantages of coal pulverization:

- Gives more surface area for a given mass of coal.
- Fast combustion as more area is exposed to heat and oxygen.
- Boiler can be started quickly.
- Fast response to load changes.



The coal is burnt in the boiler and the ash produced after the complete combustion of coal is removed to the ash handling plant and the delivered to the ash storage plant for disposal. Indian coal has ash content of 20 to 40%.

In thermal power plant bituminous coal is used because, it has low ash content.

Eg: Semi – bituminous coal is more preferable.

(b) Steam Generator (or) Boiler:

Boiler is a device. Where in water is converted into steam by utilizing the heat of combustion

This is two types:

- (i) Water tube boiler (ii) Fire tube boiler

(i) Water tube boiler:

The water flows through the tubes and hot combustion gases flow over these tube, water tube boiler are used universally for such plant.

Features of water tube boiler.

- i) Method of water circulation
- ii) Improved method of heating
- iii) Easy removal of scale from inside the tube.

(ii) Fire Tube Boiler:

The hot combustion gases flow through the tube. Water is surrounded by the fire tube.

(c) Super Heater:

A super heater is a device which raises the temperature of the steam much above the boiling point of water. Super heater heat is taken from flue gases. Normal radiant type of heat transfer method is used. It is placed between boiler and turbine. It is improved efficiency of plant.

The use of Super heater:

- (a) Reduce the steam consumption per given output.
- (b) Reduce the condensation loss.
- (c) Eliminates the erosion of steam turbine blades due to the absence of moisture.
- (d) Improves the overall efficiency.

(d) Economizer:

Economizer mainly consists of closely spaced parallel tubes through which feed water on its way to boiler flows and the fuel gases flows outside the tubes. Some of the heat energy of the fuel gases is recovered by heating the feed water near saturation but, not converted into steam this results saving in fuel (10-25%) and raising boiler efficiency.

(e) Air preheater:

An Airpreheater increases the temperature of the air supplied for coal burning by deriving heat from fuel gases. Air is drawn from the atmosphere. By a forced draught fan and is passed through air preheated before supplying to the boiler furnace. Air pre heaters are placed in between economizer and chimney. The efficiency of boiler is increased.

(f) Steam Turbine:

A steam turbine may be defined as a rotating machine, which converts the energy contained in steam into mechanical energy or rotary energy.

Basically there are two types of Turbines

- i) Impulse Turbine
- ii) Reaction turbine
- iii) Impulse Reaction Turbine (this turbine is a combination of impulse and reaction turbines)

(i) Impulse Turbine:

The steam expands completely in the stationary nozzles (or fixed blades), the pressure over the moving blades remaining constant, so the steam attains a high velocity and impinges against the moving blades this results in the impulsive force on the moving blades which sets the rotor rotating.

(ii) Reaction turbine:

The steam is partially expanded in the stationary nozzles, the remaining expansion takes place during its flow over the moving blades, and the result is that the momentum of the steam causes a reaction force on the moving blades which sets the rotor in motion.



(g) Condenser:

A condenser is a device which condenses the steam at the exhaust of Turbine. It serves two important functions. (1) It creates a very low pressure at the exhaust of Turbine, thus permitting expansion of the steam in the prime mover to a very low pressure. This helps in converting heat energy of steam into Mechanical energy in the prime mover. (2) The condensed steam can be used as feed water to the boiler. It improves power plant efficiency.

(h) Cooling Towers:

A condenser is a device in which exhaust steam from steam turbine is condensed and the heat energy given by the steam during condensation is taken up by the cooling tower.

Different types of cooling Towers are:

1. Natural draught cooling towers
2. Forced draught cooling towers
3. Induced draught cooling towers.

(i) Feed Water Heater:

The steam coming out of the turbine is condensed and the condensate is fed back to the boiler as feed water.

It is necessary to heat the feed water before feeding it to the boiler for the following reasons:

1. Feed water heating improves the overall efficiency of the plant.
2. The dissolved oxygen and carbon dioxide are removed in the feed water heater.
3. The quantity of steam produced by the boiler is increased.
4. Thermal stresses due to cold water entering the boiler are eliminated.

(j) Generator:

- Each alternator is coupled to a steam turbine and converts mechanical energy of the turbine into electrical energy.
- Alternator used in thermal power plants are of usually 2 – pole (or 4 – pole) running at

3000 rpm (or 1500 rpm) for 50 Hz operation.

(k) Exciter:

- Exciters are nothing but the D.C. Generators their main function is to supply dc power to the field system (rotor).
- Exciters are mounted on the same shaft of the alternator.
- The capacity of the exciter is about 0.5% to 3% of the main alternator capacity.
- In some cases a pilot exciter may be used to excite the main exciter itself.

(l) Electro static precipitator: The use of electrostatic precipitator is to remove fine, dust particles from flue gas. It is connected to high D.C. voltage about 30 kV. It is placed between combustion chamber and chimney

4. Advantages of Steam Power Plants:

- (a) Power stations can be located near the load centers.
- (b) It requires less space as compared to the hydro – electric power station.
- (c) Requires less transmission and distribution.
- (d) Long summer will not affect the power generation like hydel power.

5. Disadvantages of Steam Power Plants:

- (a) It pollutes the atmosphere due to the production of large amount of smoke and fumes
- (b) It is costlier in running cost as compared to hydro-electric plant.

6. Efficiency of Steam Power Station:

The overall efficiency of a steam power is quite low (about 29%) due to mainly two reasons.

- (a) Huge amount of heat is lost in the condenser.
- (b) Heat losses occur at various stages of the plant.

Thermal efficiency: The ratio of heat equivalent of mechanical energy transmitted to the turbine shaft to the heat of combustion of coal is known as thermal efficiency of steam power station.

Thermal efficiency η_{thermal}

$$= \frac{\text{heat equivalent of mech. energy transmitted to turbine shaft}}{\text{heat of coal combustion}}$$

Overall efficiency: The ratio of heat equivalent of electrical output to the heat of combustion of coal is known as overall efficiency of steam power station.

Overall efficiency

$$= \frac{\text{heat equivalent of electrical output}}{\text{heat of combustion of coal}}$$

Overall efficiency

$$= \text{thermal efficiency} \times \text{electrical efficiency}$$

1.2 Hydro Electric Power Station

A generating station which utilizes the potential energy of water at a high level for the generation of electrical energy is known as a hydroelectric power station.

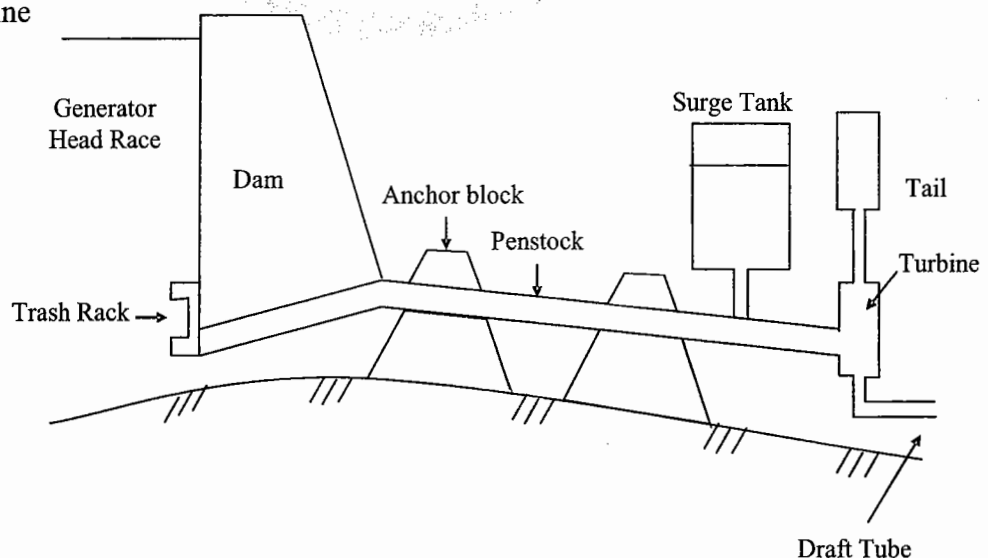
1. Selection of site for Hydro Electric Station:

- Sufficient quantity of water at a reasonable head should be available.
- More area is required for reservoir and dam.
- It should allow strong foundation with low cost.
- There should not be any leakage of water in future.
- The selected site should have large catchments area, to maintain a certain water level.
- Distance of power station site from load centers must not be longer. This is to minimize to transmission cost
- Construction materials shall be available very nearby.
- Submergence area shall be minimum

2 Schematic Arrangement of Hydroelectric Power Station:

Main elements in hydro plant

- 1) Hydraulic Structure
- 2) Water turbine
- 3) Alternator





1. Hydraulic Structure: Hydraulic structures in a hydro electric power station include as.

- (a) Water Reservoir
- (b) Dam
- (c) Trash Rack
- (d) Fore bay
- (e) Surge tank
- (f) Penstock
- (g) Spill way

(a) Water Reservoir:

- It is a basic requirement of a hydro – electric plant. Its purpose is to store water which may be utilized to run the prime mover to produce electrical power.
- A reservoir stores water during the rainy season and supplies the same during the dry season.

(b) Dam:

- The function of dam is to provide a head of water to be utilized in the water turbine.
- It helps to increase the working head of the power plant.

Intake Structure: It is an arrangement to supply water into a water conductor system.

(c) Trash Rack:

- The purpose of providing a trash rack is to prevent entering of debris which might damage the wicket gates and turbines runners.
- Trash rack is made of steel bars and is placed across the intake at the beginning of water conductor system (i.e., at the beginning of penstock).

(d) Fore Bay:

- It is an enlarged body of water just upstream of the penstocks.
- The fore bay serves as a regulatory reservoir storing water temporarily when load on the plant is reduced and to supply extra water required when load increases suddenly.

- It distributes river water to various penstocks. It is also known as **head pond**.

(e) Surge Tank:

- Surge tanks are large cylindrical tanks provided near the downstream end of the long penstock.
- Usually provided in high head and medium head plants, between dam and power house.

Function of surge tank:

- To provide better regulation of flow of water when the load on the system fluctuates. i.e., stores water when the load decreases suddenly and supplies extra water to the penstock when load increases suddenly.
- Surge tank controls the pressure variations resulting from rapid variations in water flow in penstock and thereby **prevents water – hammer effects**.

(f) Penstock:

- The pipe between surge tank and turbine is called penstock.
- It supplies water to the turbine from surge tank or fore bay.
- Its made of steel or concrete and designed as a pressure flow pipe.
- It is supported on **anchor blocks**

(g) Spill Way :

- This may be considered as a safety valve for a dam.
- A spill way serves to discharge excess water from the reservoir.

2. Water Turbines:

Water turbines are used to convert the energy of falling water into mechanical energy.

Classification based on direction of flow:

1. Axial flow
2. Tangential flow
3. Radial flow (not used)

4. Mixed Flow (or) radial in ward flow

Axial flow turbine: In this turbine, flow is parallel to the axis of rotation of the runner.

Eg: Propeller and Kaplan turbines

Tangential flow turbine: In this, flow is tangential to the path of rotation of the runner.

Eg: Pelton turbine.

Mixed Flow turbine: In this, water enters the runner at the outer periphery in the radial direction and leaves it at the centre in axial direction.

Eg: Modern Francis turbine.

Classification based on action of water:

1. Impulse turbine

2. Reaction turbine

Impulse Turbine: In this turbine, all the available energy of water is converted into kinetic energy or velocity head by passing through a nozzle provided at the end of penstock.

Eg: Pelton wheel turbine.

Reaction turbine: In this turbine, available energy of water remains mostly as pressure energy. As the water flows through the runner, the pressure energy gradually decreases. Therefore, the runner must be enclosed in an air tight casing.

Eg: Francis, Propeller, kaplan turbines.

Specific speed of turbines:

It is defined as hypothetical speed of the turbine at which it produces 1 H.P of power under head of 1 meter. It is an important parameter in the design of turbines.

Specific speed,

$$N_s = \frac{N\sqrt{P}}{H^{5/4}} \text{ r.p.m}$$

Where, N = Actual rotation speed in rpm,

H = head in meters,

P = power in HP(= 735.5 watts)

$$N_s = (1.165) \frac{N\sqrt{P}}{H^{5/4}} \text{ r.p.m.}$$

Where P = power expressed in kW

Selection of Turbines:

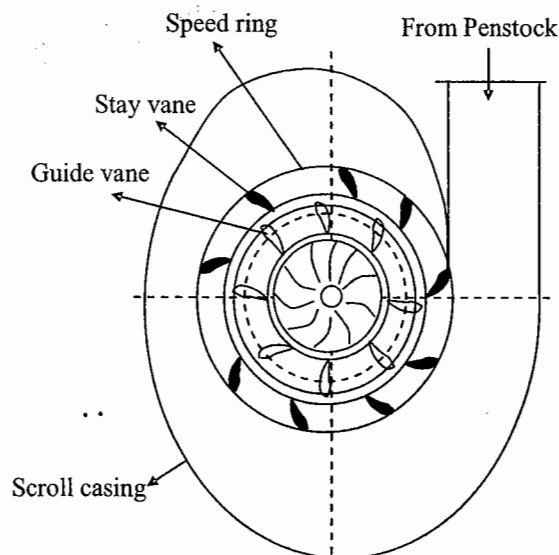
It depends on head, specific speed, efficiency, part load operation

Pelton Wheel	Francis	Propeller and Kaplan
Slow speed (12 – 70)	Medium speed (80–420)	High speed (310-1000)
Low head	Less than 15 m	Propeller or Kaplan
Medium	15-70 m	Kaplan or Francis
High head	71-250 m	Francis or pelton
Very high head	250 m and above	Pelton wheel

Parts of a Francis Turbine:

Spiral casing: The purpose of the casing is to provide an even distribution of water around the circumference of the turbine runner.

- The cross-sectional area of the casing is gradually decreased to keep the velocity of water constant throughout its path.





Speed ring or Stay ring: The purpose of ring is to direct the water from the spiral casing to the guide vanes or wicket gates.

Guide Vanes or Wicket Gates: The function is to regulate the quantity of the water supplied to the runner and to direct water on to the runner at an angle appropriate to the design.

The Guide Vanes are operated by a **Governor**.

Runner: It consists of a series of curved vanes around the circumference to change the direction of flow of water when it passes through the runner. The runner is keyed to a shaft.

Draft Tube: It is a pipe or passage of gradually increasing cross-sectional area which connects the runner exit to the tail race. The lower end must be submerged below the level of water in the tail race.

Function of Draft Tube:

- (i) It permits a negative or suction head to be established at the runner exit, thus making it possible to install the turbine above the tail race level without loss of head.
- (ii) It converts a large proportion of velocity energy rejected from the runner into useful pressure energy i.e., it acts as a recuperation of pressure energy.

GOVERNING OF TURBINES:

The operation of regulation of speed of turbine runner is known as **Governing of Turbines**. It is done automatically by means of **Governor**.

Runaway Speed : This is the maximum speed at which a turbine wheel would run under the worst conditions of operation with all gates open so as to allow all possible water inflow under maximum head.

3. Alternator:

Hydroelectric generators are low speed machines of the salient pole type, having a large number of poles, a large diameter and a short rotor (in length). The speed of a

hydroelectric generator depends on the speed of the turbine driving it, which in turn depends on the specific speed of the particular type of turbine. Generally, Generation of voltages: 3.3 kV, 5 kV and 11 kV at 50 Hz.

4. Classification of Hydro Electric Power Stations:

A. Classification based on load characteristics:

1. Base-load plants
2. Peak-load plants.

B. Classification Based on storage characteristics:

1. Run-off-river plants
2. Run-off river plants with pondage.
3. Storage plants
4. Pumped-storage plants

C. Classification based on head

1. Low head plants
2. Medium head plants
3. High head plants
4. Very High Head Plants

D. Classification based on plant capacity:

1. Micro-hydel plants
2. Mini-Hydel plants
3. Small-Hydel plants
4. Medium Capacity plants
5. High Capacity plants
6. Super Hydro plants

A. Classification based on load characteristics:

1. Base Load Plants:

- It is a plant designed to take care of based load of the grid.
- The plant operated on the large portion of the load curve.
- It supplies continuous power to the grid through out the year.
- Generally it is of large capacity.
- Its load factor is high.



2. Peak Load Plants:

- It is a plant designed to take care of peak load of the grid system.
- It operates only during the period of peak load. Hence short period of total operating time.
- Pumped storage plants are usually designed as peak load plants.

B. On the basis of hydraulic or storage consideration:

1. Run – off the river plant:

- The plant uses the water as it comes in the river. This type of power plants has no control over the river flow. It does not store the water.
- During rainy season high water flow is available and if the power plant is not able to use this large flow of water some quantity of water is allowed to flow over dam spillways as waste.
- During dry season, the power produced by such plants will be low, due to low flow rates. The utility of these plants is very less compared to other plants due to uncertainty of power supply.

2. Run – off river plants with pondage:

The usefulness of the run – off river plant is increased by providing a pond (small storage) at the plant. The pond permits to store water during off peak hours and uses during peak hours of the same day. To create pondage a barrage or a small dam may be constructed across the river.

3. Storage Plants:

- A dam is constructed across the river to create a large reservoir for storing water during rainy season and the water is released when ever required.
- The advantages of this plant is that the power generated by the plant during dry season will not be affected.

Ex: Nagarjunasagar Hydro electric plant, Srisaillam Hydro electric plant, Sileru Hydro plant etc.

4. Pumped storage plant:

Pumped storage plant in combination with hydroelectric power plant is used for supplying the sudden peak load of short duration. During the off –peak period, water is pumped back to the head water pool from the tail water pool. During the peak demand the water is drawn from the head water pool to generate power to meet the peak demand.

Pumped storage plant uses turbine called “reversible turbine”

Eg: Nagarjunasagar v Hydro electric plant and Srisaillam left bank power project.

C. Classification based on Head:

1. Low head plants:

- When the operating head is less than 15 meter the plant is named as low head plant. Run-off river plants are usually low head plants.
- This type of plant uses Kaplan Turbine.

2. Medium head plants:

- When the operating head of water is from 15 to 70 meter
- This type of plant uses Francis turbine

3. High head plants:

- When the head of water from 71 to 250 meter the plant is known as high head power plant.
- This type of plant uses Francis or Pelton turbines.

4. Very High Head plants:

- When head is more than 250 meter.
- This type of plant uses Pelton turbines.

Another classification

Low head < 30m

Medium head 30 – 250 m

High head > 250m



5. Calculation of Hydro Electric Potential:

Water head: The difference of water level is called the water head.

Gross head: The total head difference between the water levels in head race (upstream side) and tailrace (downstream side) is called as gross head or total head.

Net head or effective head: Gross head – Head loss in the Conveyor system from Head race to the entrance of turbine due to friction

- It is equal to the difference of total head at the point of entry and at the point of exit of the turbine.

Rated head: Head utilized in doing work on the turbine is called the rated head.

Rated head = Net head – Loss in guide passage and entrance of the turbine

Water Power equation (or) output equation:

Output Power, $P = w \cdot Q \cdot H \cdot \eta$ kW

where, w = specific weight of water
= 9.81 k/m^3 ,

H = net head of water in meter on the turbine

Q = quantity of water in m^3/sec .

η = over all efficiency of the system.

Output Power, $P = \frac{w \cdot Q \cdot H \cdot \eta}{75}$ h.p.

where, w = specific weight of water
= 1000 kg/m^3

Hydrograph: It is a graph between discharge and time in chronological order.

Mass curve: It is a plot of cumulative volume of water that can be stored from stream flow versus time.

Flow duration curve: It is a graph between the discharge and the percent of time the discharge equaled or exceeded.

Power duration curve: The power – duration curve is similar to that of flow duration curve but

it represents the power developed in KW or H.P against the percentage of time.

Advantages of Hydroelectric power stations:

- 1) No cost of fuel
- 2) Low maintenance cost
- 3) High plant efficiency
- 4) Plant is free from pollution
- 5) Used as multi-purpose projects (irrigation, flood control etc)
- 6) Cost per unit is less.
- 7) Suitable for variable heads and to act as a peak load plant.

Disadvantages of Hydroelectric power station:

1. The area required is more
2. High initial cost.
3. Located in remote area and require more cost on Transmission lines.

1.3 Nuclear Power Plants

NUCLEAR POWER PLANTS: A generating station in which nuclear energy is converted into electrical energy is known as a nuclear power station.

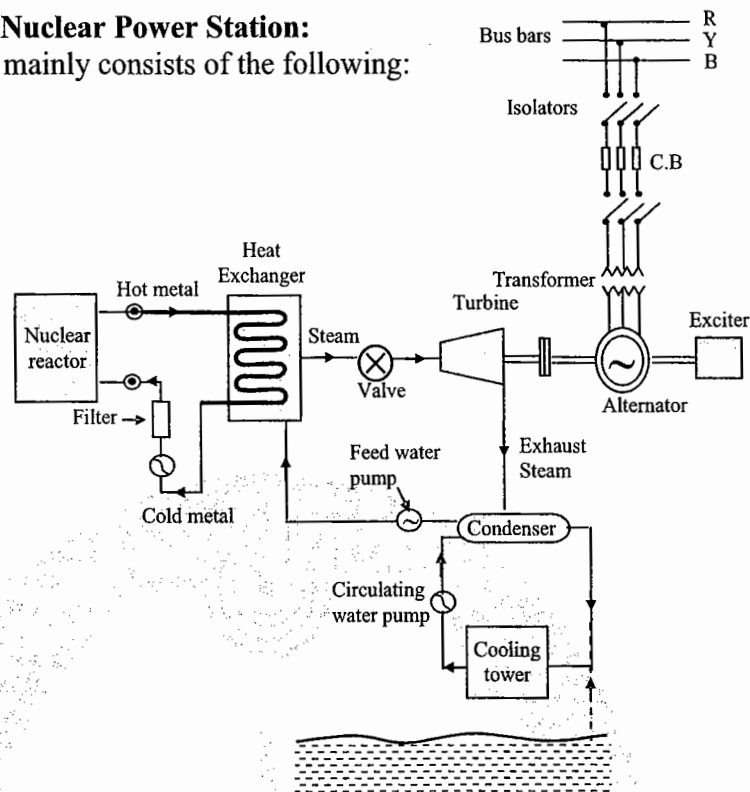
1. Selection of site for nuclear power station:

- 1) **Availability of Water:** As sufficient water is required for cooling purposes.
- 2) **Disposal of Waste:** The waste produced by fission in a nuclear power station is generally radio active which must be disposed off properly to avoid health hazards.
- 3) **Distance from populated areas:** The site selected for a nuclear power station should be quite away from the populated areas as there is a danger of presence of radio activity in the atmosphere near the plant.

2. Schematic arrangement of Nuclear Power Station:

The Nuclear Power Station mainly consists of the following:

1. Nuclear Reactor
2. Heat Exchanger
3. Steam Turbine
4. Condenser
5. Alternator



1. NUCLEAR REACTOR: It is an apparatus in which nuclear fuel (U^{235}) is subjected to nuclear fission. It controls the chain reaction that starts once the fission is done. If the chain reaction is not controlled, the result will be an explosion due to the fast increase in the energy released.

Classification of Reactors:

Reactor of a nuclear power similar to the furnace of a steam power plant. The heat liberated in the reactor due to the nuclear fission of the fuel is taken up by the coolant circulating through the reactor core. The nuclear reactor mainly consists of Fuel, core, moderator shield, reflector, reactor vessel, control rods and coolant.

(i) Types of Fuel Used:

- (a) **Natural Uranium:** Natural uranium is used as fuel and heavy water or graphite is used as moderator.
- (b) **Enriched Uranium:** The uranium contains 5 to 10% of U_{235} and ordinary water can be used as moderator.

(ii) Type of Moderator used :

- (a) Graphite reactors
- (b) Beryllium reactors
- (c) Water (ordinary or heavy water) reactors

(iii) Type of Coolant used :

- (a) Gas cooled reactors (Hydrogen, CO_2 , air)
- (b) Water (ordinary or heavy water) cooled reactors.
- (c) Liquid metal cooled reactors.
- (d) Organic liquid cooled reactors.

(iv) Types of Core:

- (a) **Homogeneous:** In this reactor fuel and moderator represent a uniform mixture such as an aqueous solution of uranium salt.
- (b) **Heterogeneous Reactor:** In such reactor fuel rods are inserted in moderator.

The fuel elements are generally arranged in some regular order forming a lattice.

Moderator: Moderator slows down neutrons before they bombard the fuel rods.



Control rods: The control rods are of cadmium or boron and are inserted into the reactor. Cadmium or Boron are strong neutron absorbers and thus regulates the supply of neutrons for fission.

Reflector: This completely surrounds the reactor core and helps to bounce escaping neutrons back into the core.

Reactor Core: This contains a number of fuel rods made of fissile material.

Types of Nuclear Reactors:

S.N	Name of the reactor	Moderator	Coolant used	Fuel
1	Advanced gas cooled reactor (AGR)	Graphite	CO ₂	Uranium
2	Pressurised water reactor (PWR)	Pressurised water	Pressurised water	Enriched Uranium Oxide
3	Boiling water Uranium Oxide	Ordinary water reactor (BWR)	Ordinary water	Enriched
4	Fast Breeder Reactor (FBR)	No moderator	Liquid sodium	U ₂₃₅ & U ₂₃₈

Fast Breeder Reactor: A fast breeder reactor is a small vessel in which the necessary quantity (corresponding to critical mass) of enriched uranium or plutonium is kept without moderator.

The vessel is surrounded by a fairly thick blanket of depleted fertile uranium. The fertile absorbs neutrons from the fissile material and gets converted into fissile material. The reactor core is cooled by liquid metal. U₂₃₅ can be converted to Pu₂₃₉; the latter can be used in other thermal or fast breeder reactor.

Breeding ratio: Ratio of number of secondary fuel atoms formed to the number of primary fuel atoms consumed. Breeding ratio equal to or greater than unity.

2. HEAT EXCHANGER: The coolant gives up heat to the heat exchanger which is utilized in raising the steam. After giving up heat, the coolant is again fed to the reactor.

3. STEAM TURBINE: The steam produced in the heat exchanger is led to the steam turbine through a valve. After doing a useful work in the turbine, the steam is exhausted to condenser.

4. CONDENSER: The condenser condenses the steam which is fed to the heat exchanger through feed water pump.

5. ALTERNATOR: The steam turbine drives the alternator which converts mechanical energy into electrical energy the output from the alternator is delivered to the bus-bars through transformer, circuit breakers and isolators.

3. Fuels used to 1690 ± 10^0 C

Boiling point 3000^0 C

Electrical resistivity $18 \mu\Omega$ - cm.

Fission: The breaking up of nuclear of heavy atoms into two nearly equal parts with release of huge amount of energy is known as nuclear fission. The release of huge amount of energy during fission is due to mass defect i.e., the mass of the final product comes out to be less than initial product. This mass defect is converted into heat energy according to Einstein relation of $E = mc^2$.

Where, E = energy;

m = mass considered (say 1 kg) ;

c = velocity of light = 3×10^8 m/sec

Fusion: The process of combination of two light nuclei to form a relatively heavier nucleus is known as fusion. The common between the two nuclear processes is that they release very large amounts of energy. But, there are many differences in the mechanism of the two processes.



1. Fission involves breaking up of a heavy nucleus into lighter nuclei. Fusion, on the other hand, involves combining of two lighter nuclei into one heavy nucleus.
2. The links of the fission process are neutrons while the links of a fusion process are protons.
3. Fission precedes best with thermal neutrons where thermal means room temperature. Fusion precedes best with thermal particles where thermal means temperatures of millions of " $^{\circ}\text{K}$ ".

Advantages of Nuclear Power Plants:

1. The amount of fuel required is quite small. Therefore, there is a considerable saving in the cost of fuel transportation.
2. A nuclear power plant requires less space as compared to any other type of the same size.

3. It has low running charges as a small amount of fuel is used for producing bulk electrical energy.
4. Running costs are less.
5. Reliable and economical for bulk generation.

Disadvantages of Nuclear Power Plants:

1. Nuclear power plants are not suitable for varying loads, as reactors cannot be easily controlled.
2. It is difficult to make the casing of the reactor, such as high temperature, neutron bombardment.
3. The disposal of the products which are radioactive is a major problem.